Industrial Machine Failure Prediction

Learn Depth Academy LLP — Track 1 Final Capstone — Problem 21

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Abstract

This project develops an educational binary classification system for predicting whether a machine is likely to experience a failure within a defined operating window. The workflow covers data inspection, exploratory analysis, preprocessing, comparison of Logistic Regression and Decision Tree models, evaluation and a lightweight Streamlit prototype. The supplied capstone brief specifies temperature, vibration, pressure, runtime and maintenance gap as expected inputs. Because the brief does not provide an assigned dataset, a reproducible synthetic dataset was created for this package.

1. Introduction

Predictive maintenance uses equipment measurements and maintenance history to identify conditions associated with potential failures. A simple machine-learning classifier provides a clear educational example of this workflow.

2. Problem Definition

The task is binary classification: predict failure=0 or failure=1 for a machine observation. The capstone brief defines the project as an educational prototype rather than a production deployment.

3. Dataset

The dataset contains 5,000 synthetic observations. Features are temperature, vibration, pressure, runtime and maintenance gap. The target is failure. A fixed random seed of 42 makes the dataset reproducible. The class counts are {0: 3842, 1: 1158}.

4. Methodology

The data are split into training and test sets using stratification. Median imputation is included in both pipelines. Logistic Regression uses standardization; the Decision Tree does not require scaling. Class weighting is enabled for both models.

5. EDA

The dataset contains 6 columns. Duplicate rows: 0. Missing values: 0. Basic descriptive statistics and target-class distribution are included in the notebook.

6. Model Development

The project compares Logistic Regression and a constrained Decision Tree. The tree uses maximum depth 5 and minimum leaf size 20 to reduce unnecessary complexity. The model selection criterion is F1-score, followed by ROC-AUC.

7. Results

Results on the held-out 20% test set are shown below. These metrics are produced by the included reproducible code.

8. Discussion

The selected model is Logistic Regression. Recall is important in this educational maintenance-risk setting because a missed failure can be more consequential than an unnecessary alert. In a real deployment, threshold selection should be based on domain-specific costs.

9. Limitations

The dataset is synthetic and therefore does not represent a particular factory, machine population or sensor system. No temporal validation, sensor drift analysis, calibration study or real maintenance-cost model is included.

10. Future Scope

Future work can use a real maintenance dataset, temporal validation, calibrated probabilities, explainability, sensor-history features and cost-sensitive threshold selection.

11. Conclusion

The project provides an end-to-end, zero-cost implementation of the assigned capstone problem using foundational ML. It includes data, notebook, trained model, evaluation, Streamlit prototype, technical paper and presentation.

References

1. Learn Depth Academy LLP, Track 1 Final Capstone, Problem 21, supplied PDF. 2. Pedregosa et al., Scikit-learn: Machine Learning in Python, Journal of Machine Learning Research, 2011.

Model comparison

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.912	0.755	0.918	0.829	0.979
Decision Tree	0.787	0.526	0.819	0.641	0.881